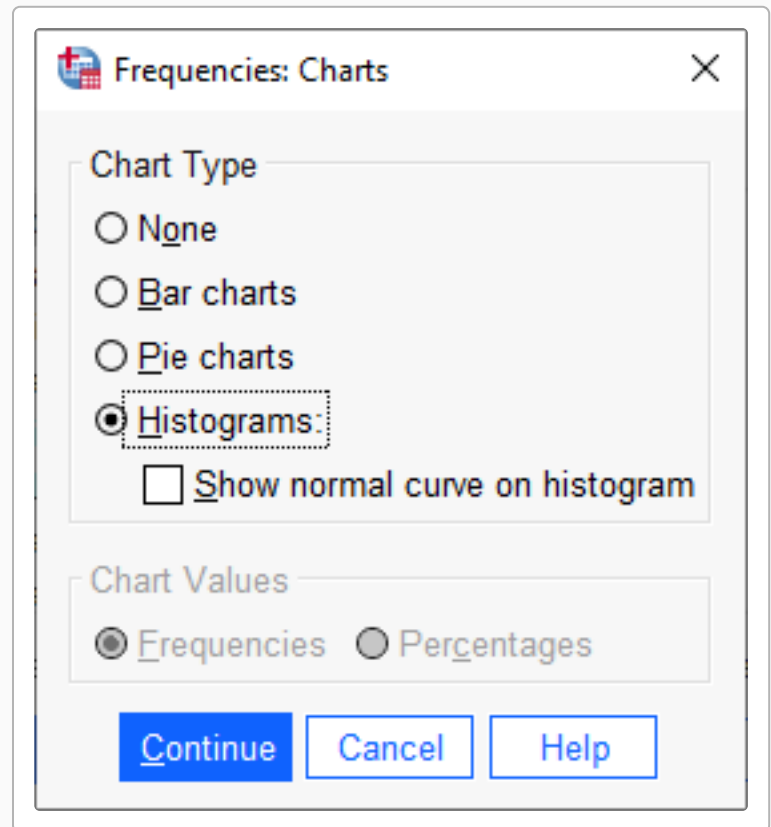
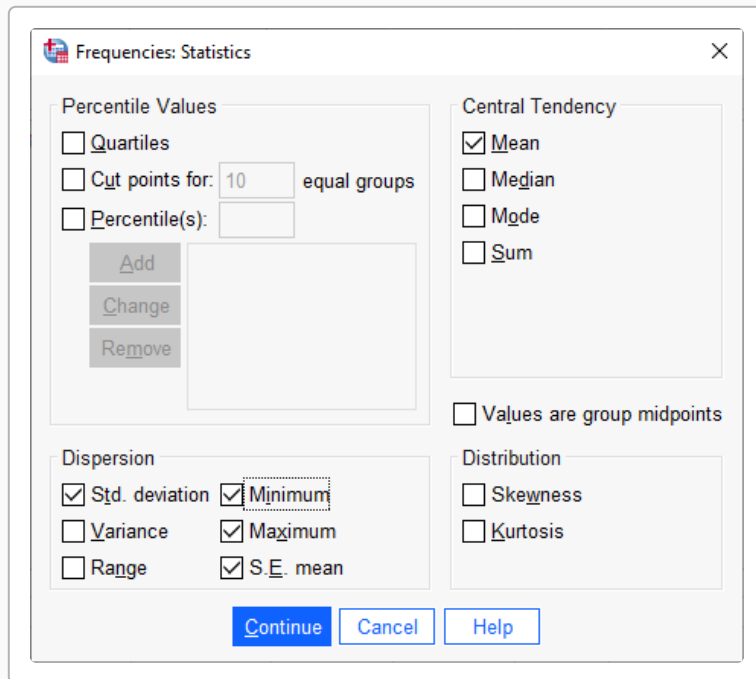
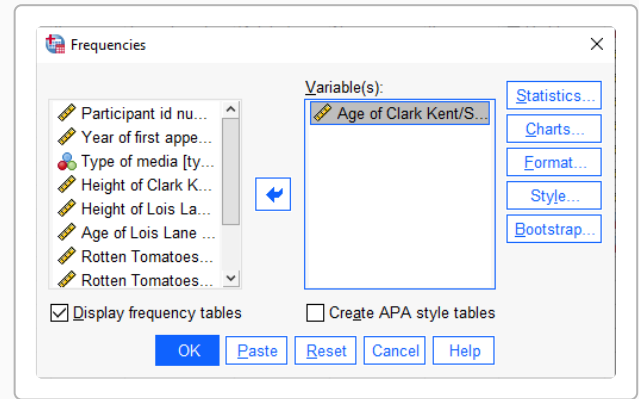


**Application:** To obtain a frequency table, a graph, and univariate statistics.

**Analyze → Descriptive Statistics → Frequencies**

- Highlight each desired variable and click the arrow button
- Click “Statistics” — indicate the univariate statistics you want computed (get mean, std dev, S.E. mean)
- Click “Charts” — click “Histogram” for quantitative variables and “Bar Chart” for qualitative variables



There are several SPSS procedures that can be used to get univariate stats and various depictions of the data. “Frequencies” is an easy starting place, because it gets some of each — graphs, frequency tables, and univariate stats.

## Univariate Stats for Quantitative Data

**Example:** Age of actor playing Clark Kent/Superman ( `clark_age` )

### Statistics

| clark_age |       |    |
|-----------|-------|----|
| N         | Valid | 11 |

Mean and standard deviation are meaningful for quantitative variables.

Combining the mean and SD tells us...

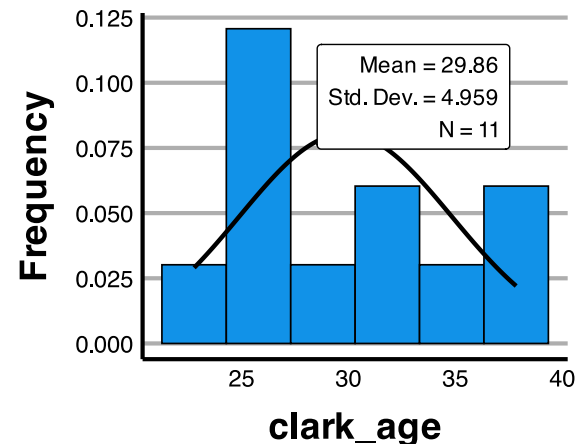
|                    |         |       |
|--------------------|---------|-------|
|                    | Missing | 0     |
| Mean               |         | 29.86 |
| Std. Error of Mean |         | 1.495 |
| Std. Deviation     |         | 4.959 |
| Minimum            |         | 22.78 |
| Maximum            |         | 37.84 |

- About 68% of Superman portrayals in this sample had actors aged between 24.8 and 34.6 years (Mean  $\pm$  1 SD)
- About 95% of the portrayals had actors aged between 19.9 and 39.5 years (Mean  $\pm$  2 SD)

## Frequency Listing & Histogram

| clark_age   |           |         |               |                    |  |
|-------------|-----------|---------|---------------|--------------------|--|
|             | Frequency | Percent | Valid Percent | Cumulative Percent |  |
| Valid 22.78 | 1         | 9.1     | 9.1           | 9.1                |  |
| 24.47       | 1         | 9.1     | 9.1           | 18.2               |  |
| 26.22       | 1         | 9.1     | 9.1           | 27.3               |  |
| 26.72       | 1         | 9.1     | 9.1           | 36.4               |  |
| 27.12       | 1         | 9.1     | 9.1           | 45.5               |  |
| 30.11       | 1         | 9.1     | 9.1           | 54.5               |  |
| 30.5        | 1         | 9.1     | 9.1           | 63.6               |  |
| 32.01       | 1         | 9.1     | 9.1           | 72.7               |  |
| 33.45       | 1         | 9.1     | 9.1           | 81.8               |  |
| 37.24       | 1         | 9.1     | 9.1           | 90.9               |  |
| 37.84       | 1         | 9.1     | 9.1           | 100.0              |  |
| Total       | 11        | 100.0   | 100.0         |                    |  |
| Total       | 11        | 100.0   |               |                    |  |

Tells the number of occurrences of each value for this variable. If there are missing values, use the “Valid Percent” column to find the percentage of cases in each value/category.



Gives a quick picture of the data. Can be used to look for noncontinuities (gaps) and skewness (asymmetry) in the distribution.

## Understanding Standard Deviation vs. Standard Error of the Mean

The **standard deviation** (SD) tells about the expected variation of individual scores around the mean of the distribution.

The **standard error of the mean** (SEM) tells about the expected variation in estimates of the population mean across multiple samples of a specific size.

We typically draw a single sample and treat the mean of that sample as our estimate of the population mean. We know that estimate is not exactly accurate, but by how much is it likely to be off?

If we took multiple samples of the same size from the same population and computed the mean for each, we could compute the standard deviation of those means as an index of the expected variability in sample estimates of the population mean.

**The standard error of the mean gives us this information, even if we have only one sample.**

How does this work? Assuming the data are normally distributed, we can estimate the variability of multiple sample means around the true population mean from the standard deviation of a single sample — the more variability there is among individuals, the more variability there is likely to be among the sample means.

## Univariate Statistics for Binary Variables

**Example:** Type of media recoded ( type2 : 1 = Film & Serial, 2 = TV Show)

### Statistics

| type2              |         |       |
|--------------------|---------|-------|
| N                  | Valid   | 11    |
|                    | Missing | 0     |
| Mean               |         | 1.45  |
| Std. Error of Mean |         | 0.157 |
| Std. Deviation     |         | 0.522 |
| Minimum            |         | 1.00  |
| Maximum            |         | 2.00  |

The decimal part of the mean tells you the proportion of scores with the higher coded value.

For a binary variable coded 1 and 2:

- Mean of 1.50 means 50% have the higher value (2)
- Mean of 1.75 means 75% have the higher value (2)
- Mean of 1.25 means 25% have the higher value (2)

The mean gives useful information for a binary variable because there is only one combination of frequencies that will yield a particular mean.

| type2 |               |           |         |               |                    |
|-------|---------------|-----------|---------|---------------|--------------------|
|       |               | Frequency | Percent | Valid Percent | Cumulative Percent |
| Valid | Film & Serial | 6         | 54.5    | 54.5          | 54.5               |
|       | TV Show       | 5         | 45.5    | 45.5          | 100.0              |
|       | Total         | 11        | 100.0   | 100.0         |                    |
| Total |               | 11        | 100.0   |               |                    |

## Univariate Statistics for Multi-Category Qualitative Variables

**Example:** Type of media ( type : 1 = Film, 2 = TV Show, 3 = Serial)

### Statistics

| type               |         |         |
|--------------------|---------|---------|
| N                  | Valid   | 11      |
|                    | Missing | 0       |
| Mean               |         |         |
| Std. Error of Mean |         |         |
| Std. Deviation     |         |         |
| Minimum            |         | Film    |
| Maximum            |         | TV Show |

**Why?** Because there are multiple combinations of frequencies that will yield the same mean.

For example, different distributions of Films, TV Shows, and Serials could produce identical means. The frequency table and bar chart are the appropriate summaries for these variables.

But remember that the mean and standard deviation are **not meaningful** for multi-category qualitative variables!

| type  |         |           |         |               |                    |
|-------|---------|-----------|---------|---------------|--------------------|
|       |         | Frequency | Percent | Valid Percent | Cumulative Percent |
| Valid | Film    | 5         | 45.5    | 45.5          | 45.5               |
|       | Serial  | 1         | 9.1     | 9.1           | 54.5               |
|       | TV Show | 5         | 45.5    | 45.5          | 100.0              |
|       | Total   | 11        | 100.0   | 100.0         |                    |
| Total |         | 11        | 100.0   |               |                    |

# SPSS Syntax

## Frequencies Syntax

### </> SPSS Syntax

```
FREQUENCIES VARIABLES=clark_age type type2  
/STATISTICS=MEAN STDDEV SEMEAN MINIMUM MAXIMUM  
/NTILES=4  
/FORMAT=NOTABLE  
/HISTOGRAM.
```

①  
②  
③  
④  
⑤

- ① List variables to analyze
- ② List statistics to compute
- ③ Get quartiles (use NTILES=2 for median only)
- ④ Omit if you want frequency tables
- ⑤ Include if you want a histogram

### ! Important

**Understanding the `EXE.` command:** If you used Compute or Recode to create a new variable before running Frequencies, be aware that SPSS creates the new variable but the values are not actually computed until either (1) you request any statistical analysis, or (2) you include and run an `EXE.` command. Running Frequencies counts as a statistical analysis, so the values will be computed at that point — but if you want to verify the new variable first, include `EXE.` after your transformation.

### i Note

#### Available Univariate Statistics:

| Type               | Statistics                                        |
|--------------------|---------------------------------------------------|
| <b>Center</b>      | MEAN, MEDIAN, MODE, SUM                           |
| <b>Variability</b> | STDDEV, VARIANCE, SEMEAN, RANGE, MINIMUM, MAXIMUM |
| <b>Shape</b>       | SKEWNESS, SESKEW, KURTOSIS, SEKURT                |